Benedikt "Ben" Safigan

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PROFESSIONAL SUMMARY

CS student at CU Boulder with hands-on, full-stack internship experience, from low-level embedded systems to React/Node.js web applications.

PROFESSIONAL EXPERIENCE

Software Engineer (Contractor) | GovWorx Jun 2026 – Present

Stack: .NET, C#, ASP.NET Core, Entity Framework Core, React, TypeScript, SQL, OpenTelemetry, Datadog

- Contributed to GovWorxDataAgent, an on-premises .NET service that captures real-time 911 emergency-services data (calls, dispatch, and location) and streams it to a cloud AI platform.
- Designed and shipped three CAD (Computer-Aided Dispatch) connectors from scratch: Mark43 REST API connector, IMC connector over Actian Zen ODBC, and CentralSquare OneSolution XML connector. Owned the schema mapping, query logic, and configuration for each.
- Extended and hardened existing connectors for public-safety vendors (Tyler, Versaterm, Hexagon, and Spillman), working across SQL Server, Oracle, Informix, ODBC, REST, and XML data sources.
- Diagnosed and fixed production issues under time pressure, including a Hexagon query failure, cursor preservation on failed pulls to prevent data loss, and clearer logging of unclean agent shutdowns.
- Shipped **50+ Linear-tracked issues across 170+ commits** through a reviewed pull-request workflow, covering backfill reliability, timezone-correctness fixes, and per-connector telemetry.
- Built a full-stack connectivity and firewall-diagnostics tool with a React and TypeScript frontend and an ASP.NET Core API, instrumented with OpenTelemetry and Datadog metrics.

Programmer & Field Technician | VARSITY FIRST Robotics Team 2996 Aug 2021 – May 2024

- Led software development for a team of 5, delivering fully functional competition robots within 6-week build cycles for 3 consecutive seasons, achieving **100% on-time delivery**.
- Wrote **2,000+ lines** of autonomous and teleoperated robot control code in Java using WPILib, achieving 95%+ complete autonomous routine success rate across 6 regional competitions with zero forfeits.

PROJECTS

Accelerated Image Processing | OpenCL, C++17, GPU Computing, Image Processing

GPU-accelerated image filter pipeline (grayscale, Gaussian blur, Sobel edge detection) in OpenCL, benchmarked against a scalar CPU baseline to achieve **15–26x speedups** on compute-bound filters.

Clockless Mutex | C++17, Distributed Systems, Concurrency

Built from scratch a distributed mutual exclusion system implementing Lamport's 1978 logical clocks paper over unreliable message-passing networks. Designed layered C++17 architecture with thread-safe components (std::mutex, std::condition_variable, RAII) and chaos injection to verify correctness despite network delays and message reordering. Demonstrates causal ordering recovery and deadlock-free operation without shared clock or central coordinator.

Spotigang | Node.js, Express, React, PostgreSQL, Spotify API

Full-stack music social feed with secure authentication, real-time Spotify API integration, and friend-sharing workflows. Deployed live on Render.

EDUCATION

University of Colorado Boulder | B.S. Computer Science (Expected May 2028)

Minor: Computer Engineering | GPA: 3.44

SKILLS

Languages: C++, Python, Java, JavaScript, TypeScript, SQL, Bash, C

Frameworks & Tools: React, Node.js, Express, .NET, ASP.NET Core, WPILib, Entity Framework Core, Docker, Git, PostgreSQL, SQL Server, OpenTelemetry, Datadog, OpenCL

Core Competencies: Full-Stack Development, System Integration, Data Pipelines, Real-Time Systems, Distributed Algorithms, Agile Development, GPU Computing, Parallel Programming